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Midterm Quiz

(Please answer all questions in English)

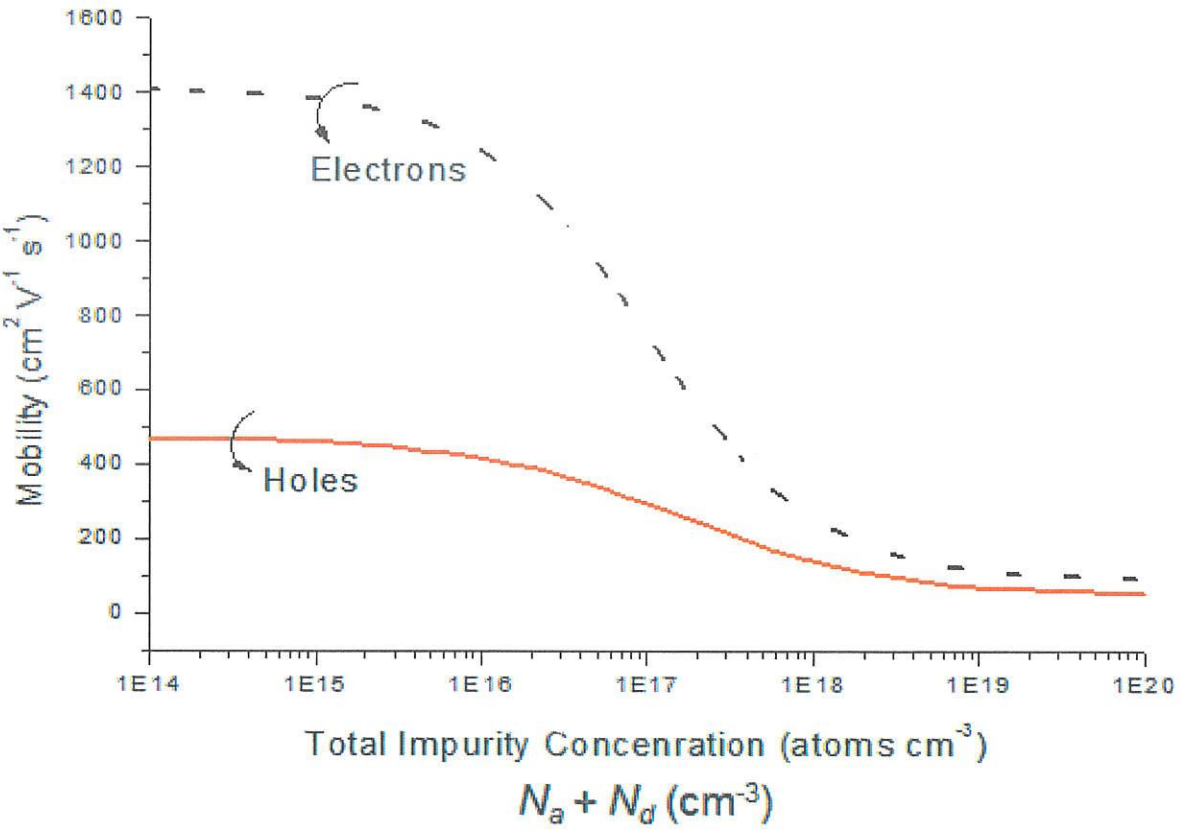
Some useful physical constants:

Electronic charge, $q = 1.6\times10^{-19}$ C
Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K
Permittivity of free space, $\varepsilon_0 = 8.854\times10^{-14}$ F/cm
Free electron mass, $m_o = 9.1\times10^{-31}$ kg
Thermal voltage kT/q = 26 mV at 300K

Properties of silicon (Si) at 300K:

Energy band gap, E_G = 1.12 eV
Intrinsic carrier concentration, $n_i = 10^{10}$ cm^{-3}
Permittivity, $\varepsilon_{Si} = 1.0\times10^{-12}$ F/cm
Effective DOS of conduction band, $N_c = 2.8\times10^{19}$ cm^{-3}
Effective DOS of valence band, $N_v = 1.04\times10^{19}$ cm^{-3}

Figure A: Carrier mobility as function of total impurity concentration at 300 K (in Si)



Question1: Please answer the following questions:

(1) What is impurity ion scattering? How is it impacted by dopant concentration and temperature? (12 points)

the conduction carrier's ~~moth~~ motion is influenced by the ~~dopant~~ doped atoms or lattice unnormal structure ~~electron/holes~~, causing the velocity decrease.

the impurity ion scattering ~~is~~ increased by the dopant concentration increases and decreases by the temperature increases.

(2) What is the Matthiessen's Rule in carrier mobility calculation? What is the basic ground for this Rule (Where does it come from)? (12 points)

① Matthiessen's Rule: $\frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{impurity}}$.

② the ~~probility~~ probability of event i happening within Δt is $\frac{1}{\Delta t}$.

and several events will happen in Δt is $\sum \frac{1}{\Delta t}$.

So, the time of carrier not being crashed ~~is~~ the probability of ② : $\frac{1}{\tau} = \frac{1}{\tau_{phonon}} + \frac{1}{\tau_{impurity}}$

because $\mu = \frac{q\tau}{m^*}$, so $\frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{impurity}}$

(3) What is the "effective mass of electron" in a semiconductor? List two factors that may have influence on its value. (12 points)

① when applied electronic field, the electron will ~~accurate~~ accelerate, but influenced by the crystalline structure and other factors, the velocity it gains will be different. we use effective mass to substitute the traditional mass, so we can conveniently calculate the electron's motion.

② Two factors: material, direction

(4) List three ways to increase carrier mobility in a semiconductor. In general, how does carrier mobility impact an electronic device's performance? (12 points)

① Three ways to increase μ: 1. decrease the impurity concentration.

2. when phonon scattering domant, decrease the temperature, whe impurity scattering domant, increase the temperature. 3. change the material to one having small effective mass.

② carrier mobility can impact the conductivity of device, and in same electron field, higher mobility leads to larger drift velocity, the device could work in high speed.

(5) What is the concept of "velocity saturation" in a semiconductor? How does it impact an electronic device's behavior? (12 points)

① Influenced by other particles, the electron in electronic field can't accelerate forever, its velocity has a maximum value.

② when we need high speed electronic device, "velocity saturation" will make it difficult to reach a high speed.

(6) What are the basic driving forces for two types of carrier movement, drift and diffusion, respectively? (12 points)

① drift: is drived by electronic field, electronic field would apply a force to charged particles.

② diffusion: is drived by thermal motion. the particles would motion in any direction, so in general, we can see they transfer from high concentration to low concentration.

(7) Regarding carrier movement in semiconductor, what is the difference of thermal velocity and drift velocity? (12 points)

① thermal velocity has random direction, but drift velocity has fixed direction which depends on electronic field's direction.

② the factors that influences them are different. thermal velocity depends on temperature and effective mass, $v_{th} = \sqrt{\frac{3kT}{m_n^*}}$. the drift velocity depends on electronic field's strength, meaning free time, effective mass, $|v_d| = qE\tau_{min}/m_n^*$

(8) What is the carrier generation process in semiconductor? List three possible mechanisms for this process. (12 points)

① carrier generation process is a process that electrons jumped (ionize) from valence band to conduction band, creating a pair of conduction electron and hole.

② three mechanisms: 1. light: electron absorbs the energy of light and ionized.
2. ionization: high energy electron loose energy with an electron ionized.
3. electronic field: which makes energy gap band thin, make it easier to generate.

(9) In a pn junction, what conditions must be met (in terms of carrier concentrations) so that (a) the n-side, and (b) the p-side can be considered as "low-level injection", respectively? (12 points)

(a) n-side: the excessive minority carrier concentration $\Delta p_n \ll n_0$,

(b) p-side: as the same, $\Delta n_p \ll p_0$.

(10) What are excessive carriers in a semiconductor? What causes them? (12 points)

① in some situations (non-thermal equilibrium), the conduction carrier will generate and become more than them at thermal equilibrium, the difference between them is excessive carriers.

② 1. Light. 2. Electronic Field. 3. Megnetic Field. 4. Mechanic Stress so on, will cause excessive carriers.

(11) In the n-side of a p^+n junction, what are "majority carrier" and "minority carrier", respectively? How about the p-side? (12 points)

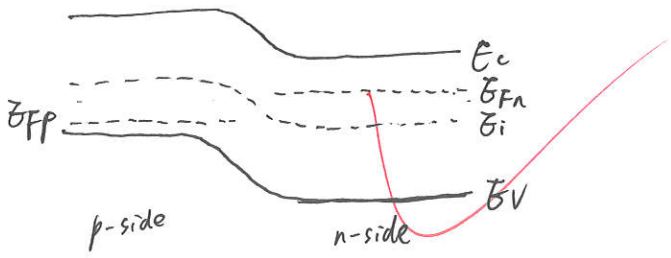
① n is majority carrier, p is minority carrier In n-side,

② In the p-side, p is majority carrier, n is minority carrier.

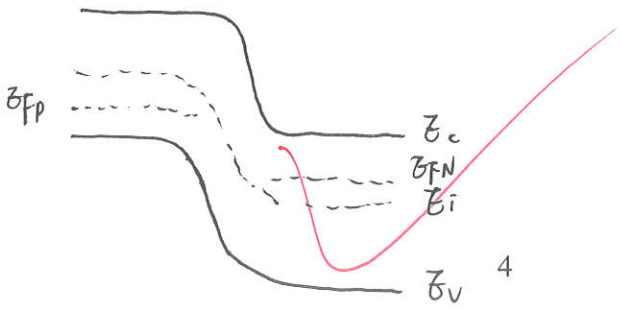
(12) What is the meaning of "quasi Fermi level" in a semiconductor? Schematically draw the band diagram of a p^+n junction in (a) forward-biased and (b) reversed-biased p^+n junction, respectively. In your plot, show E_c, E_{Fn}, E_{Fp}, E_i and E_v, respectively. (12 points)

① at non-thermal equilibrium, the pn no longer satisfys "$pn = n_i^2$", and we can't calculate them by Fermi-level. so we use quasi Fermi level, by using F_P and F_N, we can straightly derive the p, n concentration.

② forward-biased:



③ reversed-biased



Question 2:

Consider a silicon pn step junction diode maintained at room temperature, with p-side and n-side dopant concentrations $N_A = 10^{16}$ cm^{-3} and $N_D = 2\times10^{16}$ cm^{-3}, respectively. (You may assume that each side is uncompensated.) The minority carrier recombination lifetimes are $\tau_n = 10^{-6}$ s and $\tau_p = 10^{-7}$ s on the p-side and n-side, respectively.

a) Calculate the minority carrier densities at the edges of the depletion region when the applied bias V_A is $(kT/q)*\ln(10^8) \cong 0.48$V. (20 points)

b) For the applied bias in part (a), derive expressions for the minority-carrier diffusion current in the quasi-neutral regions. Sketch the electron and hole current density components (J_N and J_P, respectively) as a function of distance on both sides of the junction. Indicate the values of J_N and J_P in the depletion region, as well as the total current density J. (20 points)

Note: The minority-carrier diffusion currents are given by the equations,

$$J_{diff,n}(x'') = qD_n \frac{d\Delta n_p(x)}{dx} = q\frac{D_n}{L_n} n_{p0}\left(e^{qV_A/kT} - 1\right)e^{-x''/L_N}$$

$$J_{diff,p}(x') = qD_p \frac{d\Delta p_n(x)}{dx} = q\frac{D_p}{L_p} p_{n0}\left(e^{qV_A/kT} - 1\right)e^{-x'/L_P}$$

where x'' is the distance from the depletion region edge into the quasi-neutral p-type region and x' is the distance from the depletion region edge into the quasi-neutral n-type region.

a) $\because N_A = 10^{16}\,cm^{-3}$ $N_D = 2\times10^{16}\,cm^{-3}$ $\therefore n_p(-x_p) = \frac{n_i^2}{N_A}\cdot e^{qV_A/kT} = \frac{10^{20}}{10^{16}}\cdot 10^8 = 10^{12}\,cm^{-3}$

$p_n(x_n) = \frac{n_i^2}{N_D}\cdot e^{qV_A/kT} = \frac{10^{20}}{2\times10^{16}}\cdot 10^8 = 5\times10^{11}\,cm^{-3}$

b) ① $\because$ Einstein relationship. $D_n = \frac{kT}{q}\cdot\mu_n$ $D_p = \frac{kT}{q}\mu_p$ $\therefore \mu_n = \frac{q\tau_n}{m_n^*}$ $\mu_p = \frac{q\tau_p}{m_p^*}$

and $\therefore \mu_n = \frac{q\tau_n}{m_n^*}$ $\therefore D_n = \frac{kT}{q}\cdot\frac{q\tau_n}{m_n^*} = \frac{kT\cdot\tau_n}{m_n^*} = \frac{0.026\cdot1.6\times10^{-19}\cdot10^{-6}}{0.26\cdot9.1\times10^{-31}} =$

② $\therefore L_n = \sqrt{D_n\tau_n} = \sqrt{\frac{kT}{q}\cdot\frac{q\mu_n}{m_n^*}\cdot\tau_n} = \sqrt{\frac{kT}{m_n^*}\cdot\tau_n} = \sqrt{\frac{0.026\times1.6\times10^{-19}}{0.26\times9.1\times10^{-31}}\cdot10^{-6}} = \sqrt{0.026\cdot\pi000\cdot10^{-6}} = 5.8\times10^{-3}$ cm

③ $L_p = \sqrt{D_p\cdot\tau_p} = \sqrt{\frac{kT}{m_p^*}\tau_p} = \sqrt{\frac{0.026\times1.6\times10^{-19}}{0.39\times9.1\times10^{-31}}\cdot10^{-7}} \approx \frac{1.1}{} = \sqrt{0.026\cdot4000\cdot10^{-7}} = 1.01\times10^{-3}$ cm

④ $n_{p0} = \frac{n_i^2}{N_A} = \frac{10^{20}}{10^{16}} = 10^4\,cm^{-3}$ $p_{n0} = \frac{n_i^2}{N_D} = \frac{10^{20}}{2\times10^{16}} = 5\times10^3\,cm^{-3}$

$\frac{D_n}{L_n} = \frac{D_n}{\sqrt{D_n\tau_n}} = \sqrt{\frac{D_n}{\tau_n}} = \sqrt{\frac{kT}{q}\frac{\mu_n}{\tau_n}}$ $\frac{D_p}{L_p} = \sqrt{\frac{kT}{q}\cdot\frac{\mu_p}{\tau_p}}$

⑤ $\therefore J_{diff,n}(x'') = q\sqrt{\frac{kT}{q}\frac{\mu_n}{\tau_n}}\cdot n_{p0}\left(e^{qV_A/kT}-1\right)e^{-x''/L_N} = 1.6\times10^{-19}\cdot\sqrt{\frac{0.026\cdot1300\cdot10^{-4}}{10^{-6}}}\cdot10^4\cdot10^8\cdot e^{-x''/5.8\times10^{-3}}$

$= 9.3\times10^{-4}\,e^{-x''/5.8\times10^{-3}}$

$\approx 0.021\,e^{-x''/0.13}$ A/cm^2

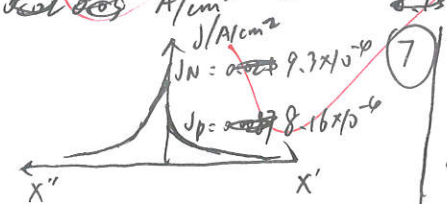
$J_{diff,p}(x') = q\sqrt{\frac{kT}{q}\frac{\mu_p}{\tau_p}}\,p_{n0}\left(e^{qV_A/kT}-1\right)e^{-x'/L_P} = 1.6\times10^{-19}\cdot\sqrt{\frac{0.026\cdot400}{10^{-7}}}\cdot5\cdot10^3\cdot10^8\,e^{-x'/1.1}$

$= 0.087\,e^{-x'/0.011}$ A/cm^2

⑥ $J = J_{PN}(x_n) + J_{NP}(x_p) \approx 1.75\times10^{-3}$ A/cm^2 $\quad 5$

$= 8.16\times10^{-4}\cdot e^{-x'/1.01\times10^{-3}}$

Answer:



⑦ $L_P = 1.01\times10^{-3}$cm $L_n = 5.8\times10^{-3}$ cm

$J_{diff,p}(x') = 8.16\times10^{-4}\,e^{-x'/(1.01\times10^{-3})}$ A/cm^2

$J_{diff,n}(x'') = 9.3\times10^{-4}\,e^{-x''/(5.8\times10^{-3})}$ A/cm^2

$J = 1.75\times10^{-3}$ A/cm^2